



EDUCATION

Program	Degree	Institution	% / CGPA	Year
Dual Degree (B.Tech + M.Tech)	Engineering Design, <i>Specialization - Robotics</i>	Indian Institute of Technology Madras	8.3/10	2024
Class XII	CBSE	Maharishi Vidya Mandir, Chetput, Chennai	95.4%	2019
Class X	CBSE	Maharishi Vidya Mandir, Chetput, Chennai	10/10	2017

- Awarded (among **30** out of 800+ students) the IIT Madras **Young Research Fellowship** based on remarkable research potential. 2021
- Secured **97.6457 Percentile** in **JEE Advanced** and **99.1046 Percentile** in **JEE Mains** out of over **1.5 Million** aspirants across India. 2019
- Secured an All India Rank of **51** in **UCEED**(Undergraduate Common Entrance Examination for Design) conducted by **IIT Bombay**. 2019

RESEARCH EXPERIENCE

SENAI Lab Research Intern Jun 2022 – Present	<i>Project aims to develop a self-balancing & autonomous bicycle under the guidance of Prof. Raghunathan Rangaswamy</i> • Developing novel momentum wheel based balancing for a bicycle to withstand 50% more impulse than usual. • Utilizing a 3D LIDAR and Stereo Camera along with odometry data through sensor fusion for 3D Localization.
Young Research Fellow IIT Madras Nov 2022 – Present	<i>An illustrious program funded by the IITM batch of '76 to support undergraduate research; Guide - Prof. Satadal Ghosh</i> • Developed a 10% faster hybrid algorithm incorporating online obstacle avoidance & path planning algorithms. • Simulated it in MATLAB environment; Drafting a research paper, to be submitted in IEEE Aerospace Journals
Human Powered Segway (Apr 2021 – Jun 2021)	<i>Led a 6 member team in Ideation of a novel design of a fully mechanical version of a Segway under Prof. Jayaganthan</i> • Improved efficiency by 67% by replacing the conventional circular pedaling system with rectilinear pedaling. • Varied the number of linkages and lengths using MATLAB to optimize the design and improve its potential.

PROFESSIONAL EXPERIENCE

Yaskawa Pvt. Ltd Robotics Intern Jun 2022 – August 2022	<i>Yaskawa Electric Pvt. Ltd. is an industry leader in Arc welding, Spot welding, Handling, Palletizing and Painting robots.</i> • Modeled and analyzed three-dimensional robotics palletizing algorithm and mechanics for mixed carton sizes. • Programmed and simulated arc welding and palletizing robots using MotoSim EG-VRC^[10] and teach pendant. • Developed a C++ algorithm to effectively use robot's memory when sending data to the robot from computer.
Blurbs Computer vision Intern Jun 2021 – Aug 2021	<i>Employs real-time vision processing techniques to assist, automate, and augment the capabilities of drones in action</i> • Analyzed and tested 7+ State of the art Multiple Object Tracking networks for the startup's first product. • Trained and tested various MOT networks like Siam-MOT and FairMOT for Surveillance with Drones and CCTV. • Used FairMOT to track and get the trail of multiple moving objects and Re-identify objects in real-time.
Avishkar Hyperloop Power Systems Engineer Oct 2020 – Jun 2021	<i>The leading Asian student competition team with the vision to make Hyperloop transportation a reality in our country</i> • Winner of the most scalable design award by Zeleros¹ and among the top 5 for mechanical and propulsion. • Worked on a custom-made Battery Management System to maintain voltage, temperature and ensure safety. • Extended battery pack life by 30% by reducing peak operational temperature by 10°C using an Al⁴ heat sink.
Imagine Pvt. Ltd. Software Dev. Intern Apr 2020 – Aug 2020	<i>Enables collaboration across different locations using VR/AR for purposes such as industrial design, support and sales.</i> • Developed a Unity3D plugin to upload any 3D model into a VR² space using Autodesk Forge APIs³ in real-time. • Built a basic Heroku app to upload and view any 3D model with 60+ file formats in the web & mobile browser • Achieved 100% reduction in staff-hours required for this specific task by automating entire process involved.

LEADERSHIP EXPERIENCE AND POSITION OF RESPONSIBILITY

Placement & Internship Cell	Department Core (May'22 - Present)	• Heading & mentoring a 3-tier team in charge of department placement and internship process 1200+ students. • Analyzed past 3 years' list of contacts to eliminate redundant calling in order to increase company conversion. • Planning & executing the logistic framework for the first ever shift to hybrid mode of the recruitment process.
	IPR Coordinator (May'21 – Apr'22)	• Member of team managing 200+ non-engineering profiles in sectors such as consulting, finance and analytics • Discovered and contacted 20+ potential recruiters based on the skill-set requirements and student requests. • Assisted the team in coordinating logistics of 100+ firms & 1000+ final year students throughout the tenure.
	ShARE'21 Junior Consultant	• Selected out of 200 applicants after multiple rounds of Global Interview comprising of consulting case analysis • Part of a global team of 700 students across 15 nations, enabling firms to transition into 'Do Well, Do Good'
	Coding and Logic Coordinator, Shaastra'21	• Handled a total footfall of 300+ teams and crafted a total of 50+ logical questions for events in Shaastra'21⁸. • Ideated and executed 2 brand new events, Mystery Rooms and E-Contest with 100+ teams participating.
	Startup Services, Associate Manager, E Cell'20	• Organized Summer Internfair 2020, handling logistics, interviews and congregating around 40+ Startups. • Organized Elevate, a fund-raising competition for startups, bringing in 150+ startups from all over the country • Managed the operational requirements of board members of unicorns throughout the Annual E-Summit

EXTRA CURRICULAR ACTIVITIES

Drawing	• Won 1 national, 5 state level and 10+ District level competitions in drawing (Oil Pastels and Pencils)
Teaching Assistant	• Assisted professors in coursework and grading of 20+ students in the courses, LifeSkills and Happiness&Success
Social Work (Gender Inequality)	• Led a 3-member team to conduct surveys to analyze the response to gender norms in visual storytelling&blogs • Brainstormed to identify the stereotypical gender inequalities and demonstrated the origin of differences. • Ideated model to ensure workplace equality by seminars, monitoring work division, and other benefits.
Game Dev. / Gaming	• Love to try out various tools and features in Unity3D; Reached a rank of Platinum 2 in Valorant by Riot Games. • Solely developed a complete 3D Game in Unity3D with Obstacles, Enemies, Points, UI and Visual Effects.
Sports	• Among the 20 out of 100 students selected for freshman Volleyball training; Part of the Hostel Volleyball team

[1] Zeleros – A Spanish company that develops Hyperloop; [2] Virtual Reality; [3] Application Programming Interface; [4] State of the Art; [5] User Interface; [6] Tool for designing and building graphical user interfaces; [8] ISO Certified Annual Technical Festival of IIT Madras; [10]; Enhanced Graphics Virtual Robot Controller